

Hyperspherical Standing-Wave Breathing Across S^3 , S^4 , and S^5

A Bessel-Projected Mechanism for Superluminal Metric Enlargement Without Local
Faster-Than-Light Signaling

Daniel Alexander Trawin
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Abstract

This manuscript formalizes the proposal that a primordial standing acoustic pattern can be made physically larger by hyperspherical metric breathing, while avoiding the incorrect claim that a local sound signal propagates faster than light. The central construction treats the observed three-dimensional cosmic shell as an S^3 projection nested inside S^4 and S^5 parent geometries. On S^n , scalar standing modes obey a dimension-shifted frequency ladder, $\omega_{n,\ell}^2 = c_s^2 \ell(\ell + n - 1)/R_n^2$, while local radial projections carry Bessel order $\nu = n/2 - 1$. Thus higher dimensions supply hidden curvature-frequency, while hyperspherical breathing enlarges the visible node spacing. The observed faster-than-light enlargement condition is not a violation of local causality, but the standard metric condition $\dot{r}_{\text{node}} = H_{\text{obs}} r_{\text{node}} > c$. The result is a disciplined version of the hypothesis: sound can seed the ruler, light can carry the snapshot, and a breathing metric can inflate the projected standing-wave geometry.

Keywords: hyperspherical breathing; S^3 ; S^4 ; S^5 ; Bessel zeros; baryon acoustic oscillations; sound horizon; metric expansion; dynamic dark energy; TZPID.

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1 Purpose and scope

The working question is whether a standing wave can be made larger by a jump in frequency, wavelength, and Bessel-zero structure in a way that resembles the faster-than-light growth associated with early-universe expansion. The strict answer is twofold.

First, a local wave in a medium does not transmit information instantaneously. A sound wave in the early photon–baryon plasma has a sound speed c_s below c , and no local signal outruns the local causal cone. Second, a metric scale factor can enlarge proper distances between comoving points faster than c without violating relativity. The clean target is therefore not “instantaneous Bessel propagation,” but rather:

Bessel or hyperspherical standing nodes fixed in comoving coordinates, enlarged by
metric breathing.

(1)

This document develops that idea for S^3 , S^4 , and S^5 , then gives the interaction map among them. It also ties the construction to the empirical acoustic-ruler frame used in the Spartan Dawn test, where the sound horizon was computed near the Planck value and dynamic breathing pressure was favored over static curvature in a compressed BAO+CMB+SNe comparison [1–3].

2 Acoustic premise: sound before free-streaming light

In standard cosmology, the early universe contained a tightly coupled photon–baryon plasma. Photons existed, but before recombination they were strongly coupled to charged matter and did not freely propagate as the cosmic microwave background. Acoustic perturbations in this plasma left the baryon acoustic oscillation (BAO) standard ruler, later observed in the clustering of galaxies [4, 5, 11].

Thus the disciplined statement is not that sound existed before photons existed. It is:

Acoustic pressure modes shaped the plasma before the oldest freely traveling light was released.

(2)

The approximate sound speed of the photon–baryon fluid is

$$c_s(a) = \frac{c}{\sqrt{3(1 + R_b(a))}}, \quad R_b(a) = \frac{3\rho_b(a)}{4\rho_\gamma(a)}. \quad (3)$$

When $R_b \ll 1$, this tends to

$$c_s \approx \frac{c}{\sqrt{3}}. \quad (4)$$

This is fast, but still subluminal. Any faster-than-light behavior in the model must therefore be assigned to the metric scale factor or to projection, not to ordinary local acoustic propagation.

3 Metric breathing and the Hubble law

Let $R(t)$ be a hyperspherical scale radius. The Hubble rate can be read as a fractional breathing rate:

$$H(t) = \frac{\dot{a}(t)}{a(t)} = \frac{\dot{R}(t)}{R(t)}. \quad (5)$$

For a comoving angular coordinate χ , the proper radial separation is

$$D(t) = R(t)\chi. \quad (6)$$

Differentiation gives

$$\dot{D}(t) = \dot{R}(t)\chi = \frac{\dot{R}(t)}{R(t)}D(t) = H(t)D(t). \quad (7)$$

Therefore recession faster than c occurs whenever

$$\boxed{D(t) > \frac{c}{H(t)}}. \quad (8)$$

This is not local faster-than-light motion through space. It is metric growth of proper distance between comoving points, as in the standard FLRW description of cosmic expansion [9, 10].

4 Geometry of S^n breathing

Definition 1 (Round n -sphere). *The round n -sphere of radius R is*

$$S^n(R) = \left\{ x \in \mathbb{R}^{n+1} : \sum_{i=1}^{n+1} x_i^2 = R^2 \right\}. \quad (9)$$

Its breathing spacetime ansatz is

$$ds^2 = -c^2 dt^2 + R_n(t)^2 d\Omega_n^2. \quad (10)$$

The n -volume of $S^n(R)$ is

$$\text{Vol}(S^n(R)) = \frac{2\pi^{(n+1)/2}}{\Gamma((n+1)/2)} R^n. \quad (11)$$

Taking a logarithmic derivative yields the universal volume-breathing identity

$$\boxed{\frac{\dot{V}_{S^n}}{V_{S^n}} = n \frac{\dot{R}_n}{R_n} = nH_n}. \quad (12)$$

For the three cases of interest:

Sphere	Volume	Logarithmic volume growth
S^3	$2\pi^2 R^3$	$\dot{V}/V = 3H$
S^4	$\frac{8\pi^2}{3} R^4$	$\dot{V}/V = 4H$
S^5	$\pi^3 R^5$	$\dot{V}/V = 5H$

At the same fractional radius-breathing rate H , higher dimension amplifies volume production. This does not by itself prove physical extra dimensions; it only states the exact geometric scaling that any S^n breathing model must obey.

5 Standing waves on S^n

Let Δ_{S^n} denote the Laplace–Beltrami operator on the unit n -sphere. Hyperspherical harmonics obey

$$\Delta_{S^n} Y_{\ell,\mu}^{(n)} = -\ell(\ell + n - 1) Y_{\ell,\mu}^{(n)}, \quad \ell = 0, 1, 2, \dots, \quad (13)$$

where μ denotes the degeneracy labels [8].

A scalar or acoustic perturbation on the breathing metric (10) has the model equation

$$\ddot{\Psi} + nH_n \dot{\Psi} - \frac{c_s^2}{R_n(t)^2} \Delta_{S^n} \Psi = 0. \quad (14)$$

Expand

$$\Psi(t, \Omega_n) = \sum_{\ell,\mu} q_{\ell\mu}(t) Y_{\ell,\mu}^{(n)}(\Omega_n). \quad (15)$$

Then each mode obeys

$$\ddot{q}_{\ell\mu} + nH_n \dot{q}_{\ell\mu} + \frac{c_s^2 \ell(\ell + n - 1)}{R_n(t)^2} q_{\ell\mu} = 0. \quad (16)$$

In the slowly varying or instantaneous-frequency approximation,

$$\omega_{n,\ell}(t) = \frac{c_s}{R_n(t)} \sqrt{\ell(\ell + n - 1)}. \quad (17)$$

For $n = 3, 4, 5$, this becomes

$$\omega_{3,\ell} = \frac{c_s}{R} \sqrt{\ell(\ell + 2)}, \quad (18)$$

$$\omega_{4,\ell} = \frac{c_s}{R} \sqrt{\ell(\ell + 3)}, \quad (19)$$

$$\omega_{5,\ell} = \frac{c_s}{R} \sqrt{\ell(\ell + 4)}. \quad (20)$$

The squared-frequency steps obey

$$\omega_{4,\ell}^2 - \omega_{3,\ell}^2 = \omega_{5,\ell}^2 - \omega_{4,\ell}^2 = \frac{c_s^2}{R^2} \ell. \quad (21)$$

For the fundamental $\ell = 1$,

$$\omega_{3,1} = \frac{c_s}{R} \sqrt{3}, \quad \omega_{4,1} = \frac{2c_s}{R}, \quad \omega_{5,1} = \frac{c_s}{R} \sqrt{5}. \quad (22)$$

Thus

$$\frac{\omega_{4,1}}{\omega_{3,1}} = \sqrt{\frac{4}{3}} \approx 1.1547, \quad \frac{\omega_{5,1}}{\omega_{3,1}} = \sqrt{\frac{5}{3}} \approx 1.2910, \quad \frac{\omega_{5,1}}{\omega_{4,1}} = \sqrt{\frac{5}{4}} \approx 1.1180. \quad (23)$$

Proposition 1 (Hidden curvature-frequency). *At fixed radius, sound speed, and mode number ℓ , lifting a standing wave from S^3 to S^4 or S^5 increases the mode frequency through the eigenvalue factor $\ell(\ell + n - 1)$. Higher spherical dimension therefore contributes hidden geometric frequency.*

Proof. Equation (17) gives $\omega_{n,\ell}^2 = c_s^2 \ell(\ell + n - 1)/R^2$. Increasing n by one adds $c_s^2 \ell/R^2$ to ω^2 at fixed R and ℓ . $\square$

6 Bessel projection and radial zeros

A projected circular standing wave in a two-dimensional plane is naturally described by $J_0(kr)$. Its first positive zero is

$$j_{0,1} \approx 2.404825577, \quad (24)$$

with notation and zero structure tabulated by the NIST Digital Library of Mathematical Functions [6].

In local flat n -dimensional radial approximation, the radial Helmholtz solution has the form

$$u_n(r) \sim r^{1-n/2} J_\nu(kr), \quad \nu = \frac{n}{2} - 1. \quad (25)$$

Thus the visible J_0 pattern may be treated as a low-dimensional shadow of higher-order radial Bessel structure.

Geometry or projection	Effective dimension	Bessel order	First radial zero
Planar shadow	2	J_0	2.4048
S^3 local radial parent	3	$J_{1/2}$	$\pi \approx 3.1416$
S^4 local radial parent	4	J_1	3.8317
S^5 local radial parent	5	$J_{3/2}$	4.4934

The S^3 entry follows from $J_{1/2}(x) \propto \sin x / \sqrt{x}$; the S^5 entry follows from the first nonzero root of the associated spherical Bessel structure, equivalently the first positive solution of $\tan x = x$ for $J_{3/2}$ -type radial behavior [7].

Proposition 2 (Outward node shift under dimensional lift). *Under the local radial approximation (25), the first radial node shifts outward as the parent dimension rises from a J_0 planar shadow through S^3 , S^4 , and S^5 .*

This supports the interpretive move: a visible J_0 ring can be the projected trace of a deeper hyperspherical radial mode. The parent geometry has a different Bessel order; the observed plane receives only its shadow.

7 Nested $S^5 \rightarrow S^4 \rightarrow S^3$ interaction

The key interaction mechanism is foliation. A four-sphere can be foliated by three-sphere slices:

$$d\Omega_4^2 = d\alpha^2 + \sin^2 \alpha \, d\Omega_3^2. \quad (26)$$

The visible S^3 radius inside S^4 is therefore

$$r_3(t) = R_4(t) \sin \alpha(t). \quad (27)$$

Taking a logarithmic derivative gives

$$\boxed{H_{3,\text{obs}} = \frac{\dot{r}_3}{r_3} = H_4 + \dot{\alpha} \cot \alpha.} \quad (28)$$

If α is fixed, the observed S^3 breathing equals parent S^4 breathing. If α drifts or twists, the observed S^3 expansion includes the extra projection term $\dot{\alpha} \cot \alpha$.

Likewise, a five-sphere can be foliated by four-sphere slices:

$$d\Omega_5^2 = d\beta^2 + \sin^2 \beta d\Omega_4^2. \quad (29)$$

Substituting (26) gives

$$d\Omega_5^2 = d\beta^2 + \sin^2 \beta \left[d\alpha^2 + \sin^2 \alpha d\Omega_3^2 \right]. \quad (30)$$

The visible S^3 radius nested inside S^5 is

$$r_3(t) = R_5(t) \sin \beta(t) \sin \alpha(t). \quad (31)$$

Therefore

$$\boxed{H_{3,\text{obs}} = H_5 + \dot{\beta} \cot \beta + \dot{\alpha} \cot \alpha.} \quad (32)$$

Proposition 3 (Projected breathing law). *For a visible S^3 slice nested inside a breathing S^5 , the observed three-dimensional Hubble-like breathing rate decomposes into parent breathing plus two angular drift terms:*

$$H_{3,\text{obs}} = H_5 + \dot{\beta} \cot \beta + \dot{\alpha} \cot \alpha. \quad (33)$$

Proof. Take the logarithmic derivative of (31). Since

$$\ln r_3 = \ln R_5 + \ln(\sin \beta) + \ln(\sin \alpha), \quad (34)$$

one obtains

$$\frac{\dot{r}_3}{r_3} = \frac{\dot{R}_5}{R_5} + \dot{\beta} \cot \beta + \dot{\alpha} \cot \alpha. \quad (35)$$

□

This is the clean mathematical form of the interaction:

$$\boxed{\text{visible breathing} = \text{parent } S^5 \text{ breathing} + S^5 \rightarrow S^4 \text{ drift} + S^4 \rightarrow S^3 \text{ drift}.} \quad (36)$$

8 Mode decomposition across the nested spheres

The S^5 Laplacian splits according to the foliation:

$$\Delta_{S^5} = \partial_\beta^2 + 4 \cot \beta \partial_\beta + \frac{1}{\sin^2 \beta} \Delta_{S^4}. \quad (37)$$

Similarly,

$$\Delta_{S^4} = \partial_\alpha^2 + 3 \cot \alpha \partial_\alpha + \frac{1}{\sin^2 \alpha} \Delta_{S^3}. \quad (38)$$

A separated S^5 mode can be organized schematically as

$$\Psi_{L,M,m}(t, \beta, \alpha, \Omega_3) = T_L(t) B_{L,M}(\beta) A_{M,m}(\alpha) Y_m^{S^3}(\Omega_3), \quad (39)$$

with mode labels ordered by

$$L \geq M \geq m. \quad (40)$$

In this language,

$$S^5 \text{ parent mode } L \rightarrow S^4 \text{ child mode } M \rightarrow S^3 \text{ visible mode } m. \quad (41)$$

The parent frequency scale is

$$\omega_{5,L} = \frac{c_s}{R_5} \sqrt{L(L+4)}, \quad (42)$$

while the visible spatial pattern is indexed by the S^3 child mode,

$$\omega_{3,m} = \frac{c_s}{r_3} \sqrt{m(m+2)}. \quad (43)$$

A lower-dimensional observer may therefore measure a large visible S^3 wavelength while the temporal variation is influenced by hidden parent index L .

9 Hidden wave number and apparent phase speed

In a local flat approximation with two hidden angular directions, the total wave number satisfies

$$k_{\text{full}}^2 = k_3^2 + k_\alpha^2 + k_\beta^2. \quad (44)$$

The full dispersion relation is

$$\omega^2 = c_s^2 (k_3^2 + k_\alpha^2 + k_\beta^2). \quad (45)$$

The visible three-dimensional wavelength is

$$\lambda_3 = \frac{2\pi}{k_3}. \quad (46)$$

A three-dimensional observer who incorrectly divides the full frequency by the visible wave number infers

$$v_{\text{app}} = \frac{\omega}{k_3} = c_s \sqrt{1 + \frac{k_\alpha^2 + k_\beta^2}{k_3^2}}. \quad (47)$$

Thus $v_{\text{app}} > c_s$ can occur as a projection effect. This is not a local superluminal signal in the full space. It is a lower-dimensional observer confusing full-dimensional timing with lower-dimensional wavelength.

Remark 1 (Wheat from chaff). *The useful statement is that hidden-dimensional wave number can make a projected wave look like a large wavelength paired with a high frequency. The unsafe statement is that the wave transmits information instantaneously. The model should keep the first and reject the second.*

10 Bessel nodes under breathing expansion

For a projected Bessel-like node with comoving wave number k , a node location can be written

$$k\chi_j = x_j, \quad (48)$$

where x_j is the relevant zero, such as $j_{0,1}$ for a planar J_0 shadow or a higher-order zero for a hyperspherical parent. The physical node radius is

$$r_j(t) = a(t)\chi_j = a(t)\frac{x_j}{k}. \quad (49)$$

Differentiation gives

$$\dot{r}_j(t) = H(t)r_j(t). \quad (50)$$

Therefore node enlargement is faster than light when

$$\boxed{H(t)r_j(t) > c.} \quad (51)$$

For the nested $S^5 \rightarrow S^3$ case, substitute (32):

$$\boxed{\left(H_5 + \dot{\beta} \cot \beta + \dot{\alpha} \cot \alpha\right) r_j(t) > c.} \quad (52)$$

This is the higher-dimensional version of superluminal inflationary enlargement. The node does not fly through the universe faster than light. Rather, the metric/projection scale that maps comoving nodes to physical radius grows fast enough that the proper separation exceeds c per unit cosmic time.

11 Frequency jump versus wavelength stretch

For an ordinary wave in a fixed medium,

$$v = f\lambda. \quad (53)$$

At fixed wave speed, increasing wavelength lowers frequency. In an expanding universe, physical wavelength scales with the scale factor:

$$\lambda_{\text{phys}}(t) \propto a(t), \quad f_{\text{phys}}(t) \propto a(t)^{-1}. \quad (54)$$

So an embedded local observer sees redshift, not a local frequency increase.

The strengthened higher-dimensional version is different:

$$\boxed{\begin{array}{c} \text{higher dimension raises hidden geometric frequency, while breathing expansion} \\ \text{stretches visible node spacing.} \end{array}} \quad (55)$$

That is, S^4 and S^5 contribute to the frequency ladder through $\ell(\ell + n - 1)$, while $R(t)$ or $a(t)$ enlarges the projected standing geometry.

If the scale factor grows exponentially,

$$a(t) = a_i \exp\left(\int_{t_i}^t H(t') dt'\right), \quad (56)$$

then node spacing grows as

$$\lambda_f = \lambda_i \exp(N_e), \quad N_e = \int_{t_i}^{t_f} H(t) dt. \quad (57)$$

This is the legitimate form of “standing wave made larger.” It is metric enlargement, not instantaneous propagation.

12 Connection to the Spartan Dawn empirical anchor

The Spartan Dawn note computed the acoustic sound horizon from the photon–baryon sound speed and compared the result to Planck-level values. It reported

$$r_s(z_*) \approx 144.5 \text{ Mpc}, \quad r_s(\text{drag}) \approx 147.1 \text{ Mpc}, \quad (58)$$

matching the usual CMB/BAO standard-ruler scale at the sub-percent level in the compressed calculation [1, 3].

The same note tested four model classes against compressed CMB acoustic scale, DESI DR2 BAO, and Pantheon+-style supernova shape information:

Model	Interpretation	Parameters	Relative result	Verdict
A	flat Λ CDM	H_0, Ω_m	benchmark	survives
B	closed Λ CDM	adds Ω_K	curvature pays little	static closure not supported
C	flat breathing $w_0 w_a$	dynamic dark energy	improved AIC	breathing favored
D	closed breathing	$\Omega_K + w_0 w_a$	best raw χ^2	curvature permitted, not forced

This aligns with the DESI DR2 report that BAO combined with CMB and supernova information gives a preference for time-evolving dark energy in the $w_0 w_a$ parameterization, with significance depending on the supernova sample [2]. The cautious conclusion is that the breathing term is empirically interesting, while closed curvature remains tightly constrained and not required.

13 TZPID interpretation

The derivation can be registered as a higher-dimensional extension of the following TZPID concepts:

Registry ID	Role in this manuscript	Mathematical expression
ID10867	Hubble-as-breathing identity	$H = \dot{R}/R, \dot{V}/V = 3H$
ID10868	Vacuum/dark-pressure reading	$\Lambda = 3\Omega_\Lambda H_0^2/c^2$ as present pressure value
ID10871	Closed breathing-enclosure Friedmann model	$H^2(a) = H_0^2[\Omega_r a^{-4} + \Omega_m a^{-3} + \Omega_K a^{-2} + \Omega_X F(a)]$
New extension	Nested hyperspherical wave projection	$H_{3,\text{obs}} = H_5 + \dot{\beta} \cot \beta + \dot{\alpha} \cot \alpha$

For dynamic pressure,

$$F(a) = \exp \left[3 \int_a^1 \frac{1 + w(a')}{a'} da' \right], \quad w(a) = w_0 + w_a(1 - a). \quad (59)$$

This is the phenomenological breathing-pressure term. A stronger theory would derive $F(a)$ from the microphysics of the enclosure rather than fit it.

14 Testable predictions and falsification channels

The proposed construction is only useful if it makes contact with observations. The main test channels are:

1. **Redshift drift.** Dynamic breathing predicts a drift

$$\dot{z} = (1+z)H_0 - H(z), \quad (60)$$

which differs from constant- Λ evolution.

2. **High-redshift BAO curvature fingerprint.** A closed term produces a $\sin \chi$ distance mapping,

$$D_M = R_c \sin \chi, \quad (61)$$

while a flat geometry uses $D_M = R_c \chi$ in the zero-curvature limit. Future high-redshift BAO measurements can reduce the allowed curvature window.

3. **Mode-ladder ratios.** If an $S^5 \rightarrow S^3$ projection leaves a detectable signature, frequency ratios associated with

$$\sqrt{\ell(\ell+4)} : \sqrt{\ell(\ell+3)} : \sqrt{\ell(\ell+2)} \quad (62)$$

may appear as structured residuals rather than arbitrary noise.

4. **Projection-driven apparent phase behavior.** Equation (47) predicts that apparent projected phase speed should vary with hidden wave-number content, not with local signal transport.

15 Conclusion

The mathematical result is clear. S^4 and S^5 do not merely make an S^3 standing wave bigger. They add hidden curvature-frequency through the hyperspherical eigenvalue ladder. The visible wavelength becomes larger only when the metric or projected radius breathes. The correct model is therefore:

higher-dimensional frequency supplies hidden timing; lower-dimensional projection supplies visible geometry; breathing supplies apparent superluminal enlargement.	(63)
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The chaff is literal instantaneous signal propagation. The wheat is metric/projection enlargement of standing nodes:

$\dot{r}_{\text{node}} = H_{3,\text{obs}} r_{\text{node}}, \quad H_{3,\text{obs}} = H_5 + \dot{\beta} \cot \beta + \dot{\alpha} \cot \alpha.$	(64)
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This keeps the inspiration of sound-before-light, Bessel-zero geometry, and faster-than-light cosmic expansion, but places each piece where the math allows it to live.

A Compact equation registry

$$\text{Vol}(S^n(R)) = \frac{2\pi^{(n+1)/2}}{\Gamma((n+1)/2)} R^n, \quad (65)$$

$$\frac{\dot{V}_{S^n}}{V_{S^n}} = nH_n, \quad (66)$$

$$\Delta_{S^n} Y_{\ell,\mu}^{(n)} = -\ell(\ell+n-1)Y_{\ell,\mu}^{(n)}, \quad (67)$$

$$\omega_{n,\ell} = \frac{c_s}{R_n} \sqrt{\ell(\ell+n-1)}, \quad (68)$$

$$u_n(r) \sim r^{1-n/2} J_{n/2-1}(kr), \quad (69)$$

$$r_3 = R_4 \sin \alpha, \quad (70)$$

$$r_3 = R_5 \sin \beta \sin \alpha, \quad (71)$$

$$H_{3,\text{obs}} = H_5 + \dot{\beta} \cot \beta + \dot{\alpha} \cot \alpha, \quad (72)$$

$$\omega^2 = c_s^2(k_3^2 + k_\alpha^2 + k_\beta^2), \quad (73)$$

$$v_{\text{app}} = c_s \sqrt{1 + \frac{k_\alpha^2 + k_\beta^2}{k_3^2}}, \quad (74)$$

$$\dot{r}_{\text{node}} = H_{3,\text{obs}} r_{\text{node}}. \quad (75)$$

B Caveat statement for peer review

This manuscript should be read as a formal hypothesis-development document, not as a completed observational proof. The higher-dimensional construction is mathematical. The empirical support is indirect: BAO confirms an acoustic standard ruler; FLRW cosmology permits metric superluminal recession; DESI DR2 motivates renewed attention to dynamical dark energy; and the uploaded Spartan Dawn note uses compressed statistics to compare static curvature against dynamic breathing. A full validation would require a Boltzmann-code likelihood analysis, explicit perturbation evolution, and a derived microphysical law for $F(a)$.

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